Docket No: _____

Invention Disclosure Form

1. Inventor(s)

a. Full Legal Name

1) Nguyen The Truong Tel. No. 0962105473 Email truong.nguyen4@datalogic.com

b. Present Address and Home Telephone Number

1) Quan Thu Duc, Ho Chi Minh city, Viet Nam

c. Citizenship

1) Vietnam

d. Business Unit

Datalogic MOB

2. Title of Invention:

Localized Peer-to-Peer Over-the-Air Distribution System for Enterprise Fleets of Android Rugged Mobile Devices

3. Conception of Invention

a. Date of First Oral Disclosure: _____

To whom: **Tran Thai, Vietnam R&D**_____

Also, identify the employer of the person to whom the invention was disclosed if the person is not a Datalogic employee

b. Date of First Drawing or Drawings _____ Present location of Drawings_____

c. Date of First Written Description _____ Present location of Description_____

d. Was invention conceived during an NPD project? ____ NO ____

If "YES", list **NPD Project Name**: _____

e. If the invention was conceived during a formal brainstorming session, list that here: _____

4. Reduction to Practice of Invention (if applicable)

a. Date Completed _____ Made by whom _____

b. Experimental Model __YES __ c. Prototype __YES__

5. Test of Invention (if applicable)

a. Date _____ b. Witnessed by:_____

c. Results _____

d. Was invention shown or disclosed outside of Datalogic? NO_____ YES _____

e. Was a confidentiality agreement executed? NO_____ YES _____

If "*YES*", please state when and with whom such confidentiality agreement was executed. _____

6. Publications

a. Has a description of the invention been published (even electronically)? NO ___ YES ____

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b. Date _____

c. Title _____

d. Name of a Publisher _____

7. Sale or Offer for Sale

a. Was product sold, offered for sale or publicly disclosed? NO _____ YES ____, Date _____

b. To whom _____

8. Public Use, Contracts, Product Implementation

a. Was invention publicly demonstrated or used? ___ YES ___ If "YES", when? __2026_____

b. Was invention conceived under Government Contract? ___NO ____

c. In which product(s) is or will invention be incorporated? Please explain the status or possibility of implementation in the product (e.g., will be included at GA, is being considered for inclusion, likely to be included in future releases, other comments, etc.).

_____

9. Signature of Inventors(s)

By signing below, the Inventor(s) agree that in exchange for good and valuable consideration, the Inventor(s) assign(s) and transfer(s) to Datalogic and all its affiliates the entire right, title and interest in and to the invention disclosed herein (which falls within my obligations under Datalogic's Employment Agreement), and any patent application(s) and letters patent(s) that may be granted for the invention in the United States of America and in any foreign countries, and in any divisions, reissues and continuations thereof, including the right to file foreign applications directly in the name of Datalogic and to claim priority rights deriving from any such United States application to which the foreign applications are entitled by virtue of international conventions.

Inventors' Signatures **Date**

1) Nguyen The Truong 01 Sep 2026

2) _____ _____

3) _____ _____

WITNESSED, READ, AND UNDERSTOOD BY: (at least one witness is required who is not an inventor)

1) Tran Hoan Thai _____

2) _____ _____

10. References

Patents or Patent Applications:

1. US 2011/0106886 A1 – "Propagating Firmware Updates In A Peer-To-Peer Network Environment," 2011.
2. US 2014/0207844 A1 – "Peer-to-peer software updates," 2014.
3. EP 1631034 A1 – "Peer-to-peer distribution of firmware," 2006.
4. US 2014/0123124 A1 – "Cloud-based firmware distribution service," 2014.

5. US 9,894,141 B2 – "Managed peer-to-peer applications, systems and methods for distributed data access and storage."

Related Publications:

6. "BitTorrent for large package distribution in the enterprise environment," IEEE International Conference on Communications (ICC), 2012. DOI: 10.1109/ICC.2012.6212681.
7. "Performance analysis of a locality-aware BitTorrent protocol in enterprise networks," Springer.
8. RFC 6763 – "DNS-Based Service Discovery," S. Cheshire, M. Krochmal, IETF, February 2013.
9. Android Developers – "Network Service Discovery (NSD)," https://developer.android.com/develop/connectivity/wifi/nsd

Competing Products:

10. SOTI MobiControl – Enterprise Mobility Management platform with OTA firmware distribution for rugged Android devices. https://www.soti.net/products/soti-mobicontrol/
11. Zebra LifeGuard Over-the-Air (OTA) – Zebra Technologies' firmware update service for enterprise Android devices. https://www.zebra.com/us/en/software/manage-your-mobile-computer/lifeguard-over-the-air.html
12. Zebra OTA Update Delivery API v3.0 – REST API for managing firmware deployment to Zebra device fleets.

11. If the Invention has been proposed to a customer or has been described to anyone outside of Datalogic, please state the circumstances and/or identify any proposal document, contract, etc.

12. If work is continuing toward building and/or testing the invention, who is doing the building and/or testing?

The inventor, Nguyen The Truong, is continuing development and testing of the software prototype.

13. When will the invention be disclosed (e.g., when will the product incorporating the invention be shipped, described to others, etc.)?

Disclosure might be tied to the product roadmap for future Datalogic Android devices (QCM4490 or QCM6690). This timeline is to be determined.

Description of Invention

General Description

The present invention is a localized peer-to-peer (P2P) Over-the-Air (OTA) distribution system that enables large firmware and operating-system update packages to propagate across an entire fleet of rugged Android mobile computers without every device independently downloading the full package from a central server. The system operates entirely within the local network perimeter and is purpose-built for the Android platform, distinguishing it fundamentally from Internet-scale P2P protocols such as BitTorrent.

A single source device (the **root seeder**) retrieves the multi-gigabyte OTA image once from the central OTA server, partitions it into thousands of fixed-size 512 KB data chunks in a single streaming pass, and simultaneously computes a per-chunk SHA-256 hash and a running whole-file SHA-256 digest. The result is a compact *InformationMetadata* descriptor that carries the total file size, chunk size, chunk count, the ordered array of per-chunk hashes, and the global file hash. This descriptor serves as the authoritative manifest for every subsequent peer transaction.

The root seeder advertises its presence on the local network and serves chunks through a lightweight embedded HTTP server. Peer devices discover the root, pull and verify the metadata, and begin fetching chunks. **Critically, the instant a peer obtains the metadata it simultaneously advertises its own sharing service and begins serving whatever chunks it already holds** — converting every downloader (leecher) into an additional distribution source (seeder). Update data therefore fans out laterally across the fleet rather than radiating from one origin, and aggregate throughput scales with the number of participating devices.

Unlike conventional BitTorrent, which relies on Internet-facing trackers, distributed hash tables (DHTs), and globally routable peers, the proposed system: (1) operates **exclusively within the enterprise local network**, confining all transfer traffic to the LAN/VLAN segment or to short-range Wi-Fi Direct links; (2) is **purpose-built for Android**, leveraging platform-native APIs (Android NSD, Wi-Fi Direct framework) rather than portable but heavyweight P2P stacks; and (3) incorporates **multicast-storm mitigation** mechanisms specifically designed for the enterprise intranet broadcast domain, a failure mode that generic P2P frameworks not only fail to address but actively exacerbate.

Two discovery-and-transport modes are selectable at deployment time through a single orchestration layer. In infrastructure-backed deployments, Android Network Service Discovery (NSD / DNS-SD) enables any device within the enterprise Wi-Fi or VLAN to locate sharing peers and a root tracker service that reports fleet-wide progress. Where no fixed infrastructure exists, Wi-Fi Direct forms a cascading tree of device groups that relay data hop-by-hop. Both modes share a common coordinator — a ShareManager driving a SeekManager (downloading) and a SeedManager (serving) — with end-to-end SHA-256 integrity verification and fault tolerance for dropped peers.

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Figure 1. System Architecture Overview

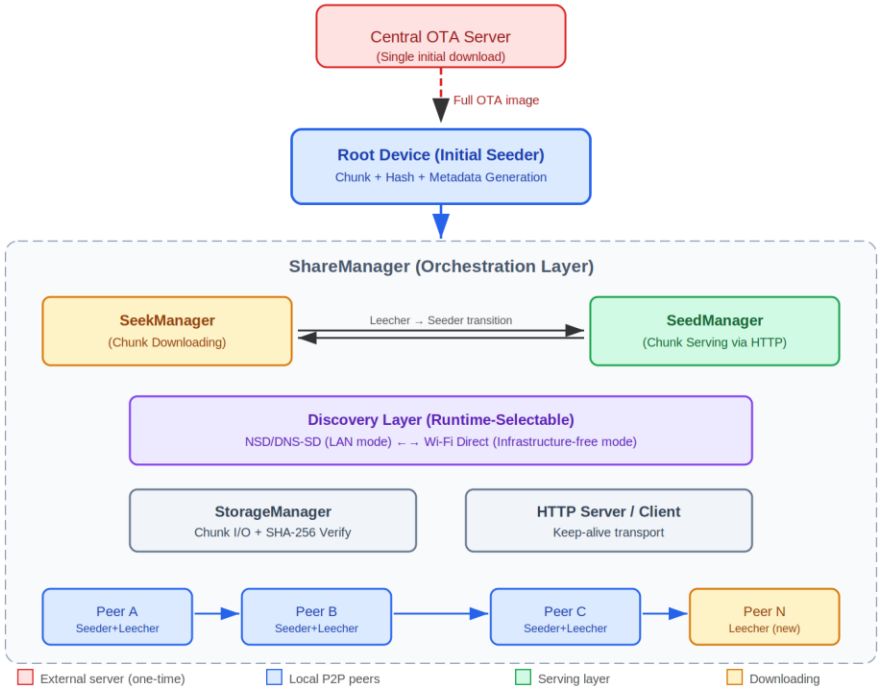


Figure 1. System Architecture Overview — ShareManager orchestrates SeekManager, SeedManager, discovery, transport, and storage layers behind a unified control plane.

Operational Problems Solved

In warehouses, factory, and distribution-center environments, hundreds to thousands of rugged devices must periodically receive the same large OTA update. Under the conventional client–server model, each device independently downloads the full multi-gigabyte image from a central server, creating three acute operational failures within the enterprise local network:

- **Core-network bandwidth saturation during concurrent mass updates:** Simultaneous full-image downloads saturate access points, uplinks, and backhaul, degrading not only the update itself but every other business-critical application sharing the enterprise network.
- **Server resource exhaustion and single point of failure:** The central OTA server becomes both a throughput bottleneck and a single point of failure as concurrent connections scale linearly with fleet size.
- **Prolonged device deployment downtime:** Finite shared bandwidth forces updates to serialize in practice, leaving devices — and the workers who depend on them — idle through extended maintenance windows.

By redistributing the transfer load onto the devices themselves and confining all traffic to the local segment (or to infrastructure-free Wi-Fi Direct links), the invention **removes the central bottleneck, collapses total fleet-wide distribution time, and eliminates the need for costly core-network hardware upgrades** to absorb periodic update events.

Figure 2. Client–Server vs. Localized P2P Distribution

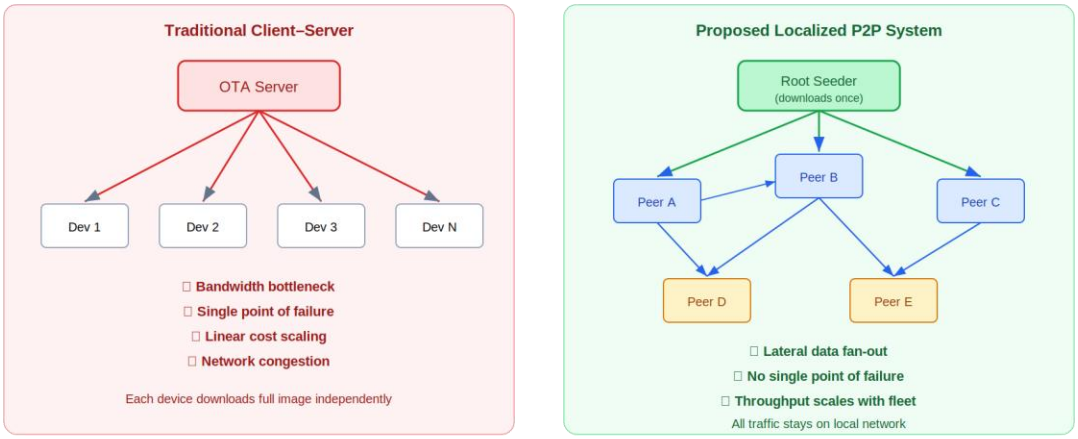


Figure 2. Client–Server vs. Localized P2P Distribution — The proposed system eliminates the central bottleneck by distributing transfer load across all participating devices within the local network.

Prior Art Background

2.1 Standard Client–Server OTA Infrastructure

The prevailing state of the art delivers OTA packages from a central server — on-premises or cloud-hosted — to which every device opens an independent connection and downloads the complete image. While simple and centrally governable, this one-to-many topology suffers from three fundamental limitations that worsen with fleet scale:

- **Server resource exhaustion.** Bandwidth, connection count, disk I/O, and CPU all grow linearly with the number of concurrent devices. Serving a multi-gigabyte image to hundreds or thousands of endpoints demands costly server or CDN over-provisioning that remains idle between update campaigns.
- **Network bottleneck.** Aggregate download demand concentrates on a single path — the WAN uplink for a cloud server, or the access-point backhaul for an on-premises one — saturating the very links shared by production traffic.
- **Single point of failure.** If the central server becomes unavailable or overloaded mid-campaign, the entire fleet stalls with no alternative data source.

2.2 General Peer-to-Peer Distribution

BitTorrent-style P2P distribution is the recognized answer to server-side scaling: peers exchange segments among themselves, so aggregate throughput grows with the swarm rather than saturating one origin.

However, general P2P frameworks are engineered for the open Internet, where peer discovery is mediated by dedicated trackers or DHTs over unicast. Transplanted naively onto an enterprise Layer-2 intranet, their continuous peer-discovery behavior becomes their dominant failure mode: **the Multicast Storm.** When hundreds of devices sharing a single VLAN/broadcast domain each run continuous multicast/mDNS-style

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discovery, the broadcast and multicast frames are replicated to every port in the segment, consuming access-point airtime and switch CPU and degrading connectivity for all hosts on the VLAN — often inflicting worse network-wide harm than the client–server congestion it was meant to relieve.

This gap — a P2P system that is both localized to the enterprise LAN and hardened against the multicast-storm hazard — is the specific void the present invention fills.

Reason for Invention

3.1 Business Value and Motivation

The invention delivers concrete, quantifiable benefits to Datalogic and to the enterprises operating its devices:

- **Substantial infrastructure cost reduction:** Because the transfer load is carried by the devices and confined to the local segment, customers no longer need to upgrade OTA servers, CDN capacity, access points, or backhaul to survive a mass-update event. Existing site infrastructure absorbs full-fleet campaigns it previously could not.
- **Maximum update speed without hardware upgrades:** Distribution time is decoupled from any single server's egress and instead scales with the number of participating peers, collapsing the maintenance window and the associated device downtime — a direct productivity gain in warehouses and factories where idle devices mean idle workers.
- **Resilience and reach:** The self-seeding swarm has no single point of failure and continues distributing even where the uplink to a central server is weak or absent (remote depots, infrastructure-free sites), extending Datalogic's serviceable deployment envelope.

3.2 Novelty and Innovation Statement

The competitive advantage and technical barrier of this invention lie in the integration of a **Technological Triad**, purpose-tuned to the constraints of rugged mobile hardware:

- **(1) Fixed-size data chunking (512 KB) with a dual-layer SHA-256 integrity protocol** — per-chunk hashing for surgical, minimal-waste re-fetch, plus a whole-file checksum for bit-exact final assurance.
- **(2) Android Network Service Discovery (NSD/DNS-SD) with role isolation and multicast-storm mitigation** — dedicated, standards-compliant service types separate the chunk-sharing role from the root/tracker role; local peer caching and duplicate-discovery suppression tame the enterprise broadcast-storm hazard.
- **(3) Wi-Fi Direct cascading tree with simultaneous store-and-forward relay** — organized as a hierarchical tree topology where each node simultaneously receives from its parent Group Owner and distributes to its own group members, enabling fully infrastructure-free distribution despite the single-radio hardware limitation.

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The integration of these three mechanisms — each individually adapted to rugged-device power, CPU, RAM, and radio constraints, and jointly orchestrated behind a unified seeker/seeder control layer — establishes a distinct intellectual-property position that is **not derivable from existing BitTorrent, IPFS, or other Internet-scale P2P protocols**, as those systems neither target local-only operation, nor address the multicast-storm problem, nor accommodate the single-radio Wi-Fi Direct constraint of mobile hardware.

3.3 Continued Development Potential

- **Integration with Datalogic's on-premises local server infrastructure:**
  Datalogic is actively deploying local OTA servers at customer factory sites. The P2P distribution engine can be integrated to complete the end-to-end OTA lifecycle:
  The local server pulls certified firmware from Datalogic's cloud, and the P2P swarm distributes it across the floor — eliminating the server-to-device bottleneck that currently limits the local server's scalability.
- **General-purpose large-file distribution beyond OTA firmware:**
  The P2P engine is payload-agnostic — APKs, configuration files, barcode libraries, or any large dataset can be distributed fleet-wide through the same mechanism.
  This gives enterprise customers a self-contained, on-device distribution channel that operates independently of third-party MDM/EMM platforms, reducing licensing costs and external dependencies..

Description

This section provides a structured architectural breakdown sufficient for a skilled engineer to understand, build, and implement the system. The system is coordinated by a central ShareManager, which drives a SeekManager (chunk downloading/leeching), a SeedManager (chunk serving), a discovery layer (NsdDiscoveryManager or WifiDirectDiscoveryManager), a lightweight HttpServer/HttpClient transport, and a StorageManager for chunk-level file I/O and verification.

4.1. Data Preparation and Metadata Initialization

The source device, having retrieved the OTA image once, partitions it into contiguous fixed-size chunks of 512 KB each (*CHUNK_SIZE = 524,288 bytes*). In a single streaming pass, the system simultaneously (a) computes a per-chunk SHA-256 hash and (b) feeds each chunk into a running whole-file SHA-256 digest. The output is an **InformationMetadata** descriptor that carries:

- The total file size of the OTA image
- The fixed chunk size (524,288 bytes)
- The total chunk count
- An ordered array of per-chunk SHA-256 hashes
- A whole-file SHA-256 hash

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This descriptor is the authoritative manifest: every peer must fetch and verify it before requesting any data chunks.

Figure 3. Data Chunking and Dual-Layer SHA-256 Verification

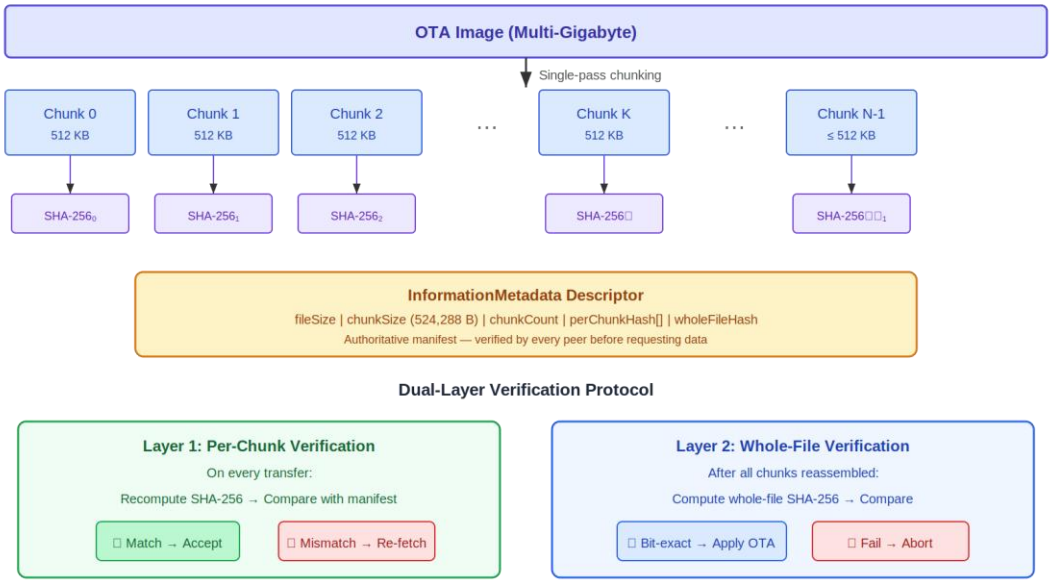


Figure 3. Data Chunking and Dual-Layer SHA-256 Verification — A single streaming pass produces per-chunk hashes and a whole-file digest. Layer 1 verifies each chunk on receipt; Layer 2 provides bit-exact assurance over the reassembled image.

Dual-Layer Verification Protocol

Layer 1 — Per-Chunk Verification (active on every transfer): Immediately upon receiving a chunk, the leecher recomputes its SHA-256 and compares it against the manifest entry. On mismatch, only that single 512 KB chunk is discarded and re-requested — corruption never forces a full re-download, minimizing bandwidth waste and blocking accidental or malicious injection of bad data into the swarm.

Layer 2 — Whole-File Verification (final assurance): After all chunks have been received and reassembled, the whole-file SHA-256 embedded in the manifest is recomputed over the complete image. This provides a bit-exact integrity gate guaranteeing the reconstructed file is byte-for-byte identical to the source package before the device is permitted to apply the update.

Position-addressed reassembly: Each chunk is written directly to its computed byte offset in a pre-sized file via RandomAccessFile/FileChannel, so chunks may arrive in any order — including non-sequentially from multiple peers — and still land at the correct position.

4.2. Infrastructure Mode — Network Service Discovery (NSD/DNS-SD)

When the enterprise LAN/VLAN is available, devices discover one another using Android Network Service Discovery over DNS-SD. This mode leverages the existing network infrastructure and requires no additional hardware or configuration beyond standard enterprise Wi-Fi.

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Service-Type Role Isolation. Role isolation is achieved not through DNS-SD subdomain nesting (which would violate RFC 6763) but through two distinct, standards-compliant service types:

- ***_largeshare._tcp*** — the chunk-sharing service that every seeding peer advertises (service name LargeShare_<deviceId>); this is the primary data-distribution channel.
- ***_largeshareprogress._tcp*** — the root/tracker service advertised exclusively by the root device, to which peers periodically POST their completion progress so an administrator can monitor fleet-wide distribution.

A seeker runs discovery for both service types. A resolved _largeshare._tcp peer becomes a leech source, while a resolved _largeshareprogress._tcp peer is retained as the root progress sink. **As soon as a seeker resolves the shared manifest, it registers its own _largeshare._tcp service, immediately re-seeding to the swarm.**

Figure 4. NSD/DNS-SD Discovery and Swarm Formation (LAN Mode)

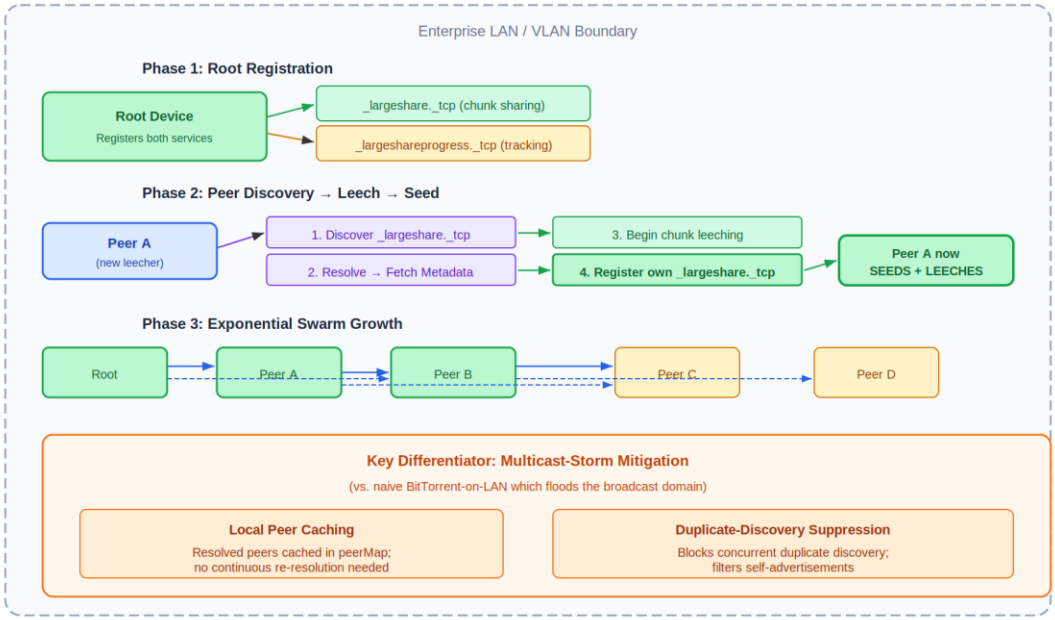


Figure 4. NSD/DNS-SD Discovery and Swarm Formation — Peers discover the root via standard service types, fetch the metadata, begin leeching, and immediately register as seeders.

Multicast-Storm Mitigation (Key Differentiator). Two code-backed mechanisms curb the discovery flooding that plagues intranet P2P:

- **Local peer caching:** Resolved peers are cached in a local peer map (ShareManager.peerMap). The system operates against this locally held peer list rather than continuously re-resolving, dramatically reducing multicast/mDNS traffic.
- **Duplicate-discovery suppression:** The discovery layer refuses to launch a second concurrent discovery for a service type already active, and filters self-advertisements, eliminating the $O(N^2)$ broadcast amplification.

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4.3. Infrastructure-Free Mode — Wi-Fi Direct Cascading Tree

Where no LAN infrastructure exists — remote depots, field deployments, or sites with damaged network equipment — the system falls back to Wi-Fi Direct short-range links organized as a cascading tree topology.

Hardware Constraint and Structural Solution. A mobile device has a single Wi-Fi radio (MAC), which means it physically cannot serve as a Wi-Fi Direct Group Owner (GO) and a Group Client simultaneously. The system resolves this through a hierarchical tree structure configured before distribution begins:

- **Level 0 (Root):** The root device acts as the Group Owner for the first Wi-Fi Direct group. It holds the complete OTA image and serves chunks to its group members.
- **Level 1..N (Intermediate Nodes):** Each device that joins a parent group as a client simultaneously creates its own Wi-Fi Direct group as GO, allowing additional devices to connect. A node therefore acts as a client of its parent's group to receive data, and simultaneously as GO of its own group to distribute data downward.

Simultaneous Store-and-Forward Relay. Unlike traditional store-and-forward where a node must complete reception before forwarding, this system implements **simultaneous receive-and-distribute**: as soon as a node receives and verifies a chunk from its parent GO, that chunk becomes immediately available to the node's own group members. Data cascades continuously down the tree without waiting for any node to complete its full download.

Figure 5. Wi-Fi Direct Cascading Tree Topology (Infrastructure-Free Mode)

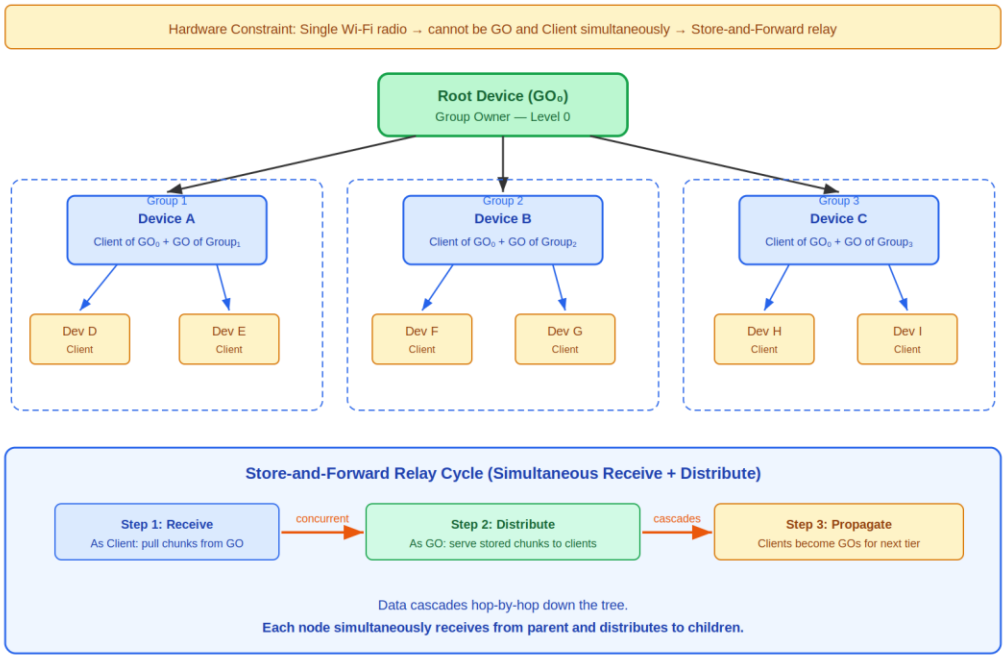


Figure 5. Wi-Fi Direct Cascading Tree Topology — Each node simultaneously receives chunks from its parent GO and distributes them to its own group members, enabling infrastructure-free distribution despite the single-radio constraint.

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4.4. Fault Tolerance and Performance Engineering

The system incorporates multiple fault-tolerance and performance-optimization mechanisms specifically tuned for the constrained CPU, RAM, and I/O budgets of rugged mobile hardware:

Peer-Drop Recovery. When a peer disappears (onPeerLost) or a transfer fails mid-chunk, the affected peer is disconnected, in-flight chunk claims are released back to the pool (compensateChunk), and ShareManager.replaceLeechingPeer promotes a different cached peer. In NSD mode, background discovery continually refreshes the candidate list.

Random Chunk Selection. The claimChunk algorithm selects a **random** chunk from {available} ∩ {not-held} ∩ {not-in-flight}, eliminating sequential contention and maximizing parallel utilization of all seeders.

Connection Pooling for Rugged Hardware. Concurrency is deliberately bound:

- Simultaneous peer leeching: max 5 connections (IN_FLIGHT_PEER_LEECHING_LIMIT = 5)
- Peer connection threads: max 3 per device (MAX_PEER_CONNECTIONS = 3)
- Socket timeouts: 5 seconds for connect/read/request

HTTP Keep-Alive Transport. The embedded HTTP server holds each TCP connection open for multiple sequential chunk requests (Connection: keep-alive), amortizing handshake overhead. Connections are torn down only on timeout, overload, or peer-list change.

Graceful Seeding Grace Period. After leeching completes, the device continues seeding for 15 seconds (STOP_SEED_DELAY_MS) before releasing its HTTP server and unregistering its discovery service, ensuring late-joining peers can still pull data.

Cross-AP Roaming and Multi-AP Environments: In enterprise deployments, devices frequently roam between different Wi-Fi Access Points as workers move across the facility floor. The proposed system handles this transparently because NSD/DNS-SD discovery operates at Layer 3 (IP network layer), not at the physical AP association layer — as long as all devices remain within the same VLAN or IP subnet (standard enterprise configuration)

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Figure 6. Fault Tolerance and Performance Engineering

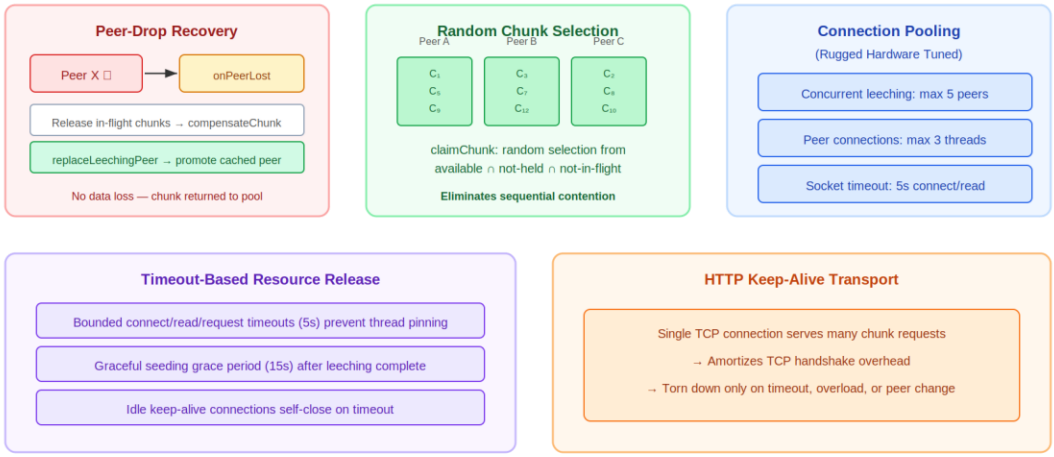


Figure 6. Fault Tolerance and Performance Engineering — Peer-drop recovery, random chunk selection, bounded connection pools, and HTTP keep-alive transport maintain distribution resilience on constrained hardware.

4.5 Theoretical Performance Estimation and Cost-Saving Analysis

The following estimation models a representative large-scale factory deployment based on published specifications of industrial-grade Wi-Fi infrastructure and Datalogic's current Android rugged device portfolio

Parameter	Value	Basis
OTA package size	2.5 GB	Typical Android full-system OTA
Fleet size	1,000 devices	Mid-to-large warehouse/factory
Device Wi-Fi capability	802.11ac, ~150 Mbps practical	Datalogic Memor 30/35, Skorpio X6 specs
Enterprise APs	30 units	Coverage for ~15,000 m² facility
AP practical throughput	~400 Mbps aggregate per radio (5 GHz, derated 50%)	
AP backhaul uplink	1 Gbps wired	Standard enterprise deploymen
Central OTA server NIC	1 Gbps	

Based on the infrastructure above, the following comparison demonstrates that the proposed P2P system delivers substantial advantages in distribution speed, bandwidth efficiency, fault resilience, and infrastructure cost avoidance over the conventional client–server model

Dimension	Client–Server	Proposed P2P
Server egress (total)	1,000 × 2.5 GB = **2,500 GB**	1 × 2.5 GB = **2.5 GB** (single seed download)
Bandwidth per device	1 Gbps ÷ 1,000 = **1 Mbps**	Peer-to-peer: **~50Mbps - 150 Mbps**
Fleet update time	2.5 GB ÷ 1 Mbps ≈ **5.6 hours**	Exponential fan-out: $\log_2(1000)$ ≈ 10 cycles × ~100s ≈ **15–20 min**
WAN/backhaul saturation	**100%** for entire duration — all production traffic degraded	**Minimal** — traffic stays local between peers
Single point of failure	**Yes** - server down = entire fleet stalled	**No** - any seeder lost, peers redistribute from remaining swarm
Server/CDN upgrade needed at scale	**Yes - 10 Gbps NIC(~$5K–$15K/site)**	**No** - scales by device count, zero additional infrastructure
Estimated time saving—~95%		**~95%**
Estimated egress saving		~99.9%

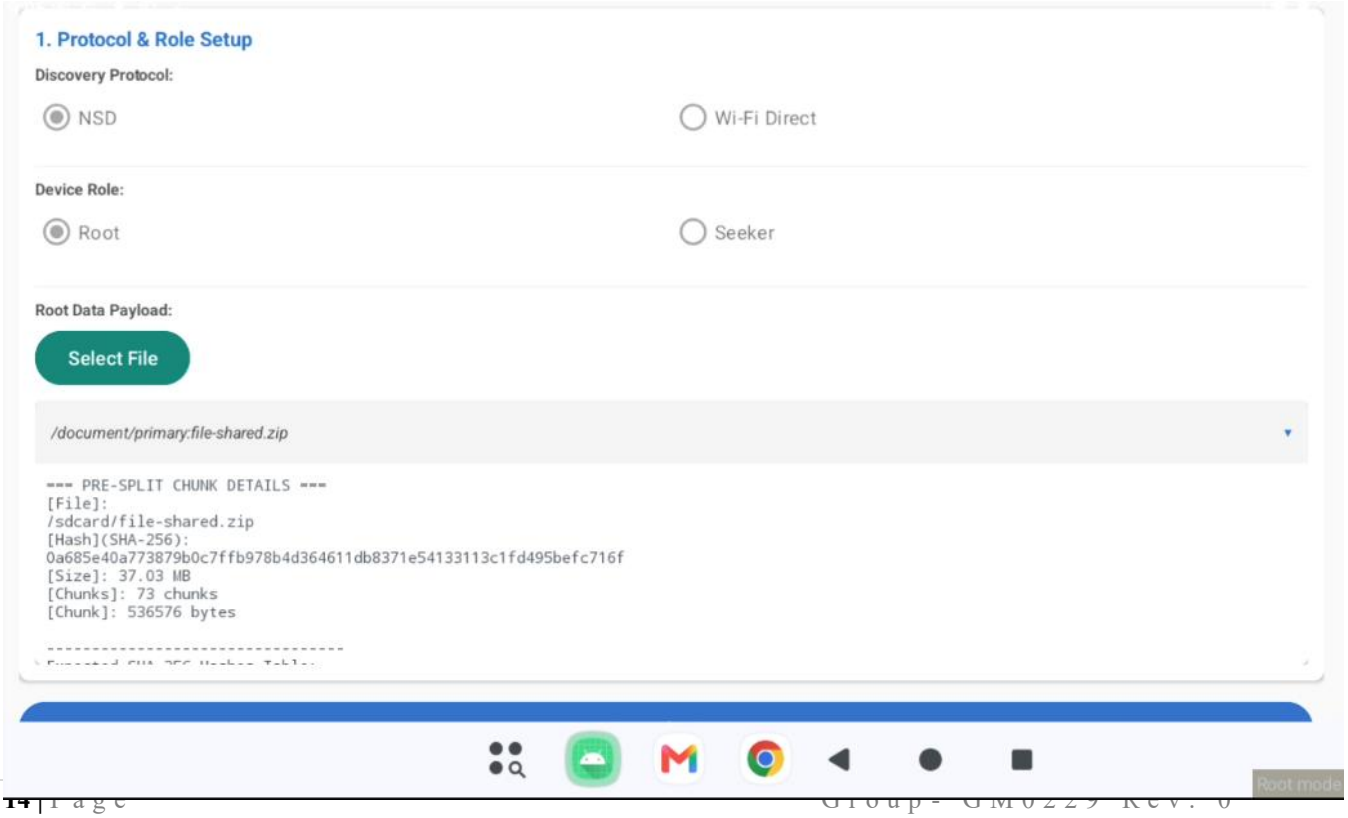
Intellectual Property Independence

The proposed system does not derive from, extend, or incorporate any component of the BitTorrent protocol specification (BEP), the IPFS/libp2p stack, or any other existing P2P distribution framework. The following architectural choices establish clear IP separation:

- **No tracker protocol:** The system uses Android NSD/DNS-SD for peer discovery — a platform-provided, standards-based mechanism — rather than a BitTorrent tracker or DHT.
- **No .torrent file or bencoding:** The metadata descriptor (InformationMetadata) is a purpose-built, application-specific manifest, not a bencoded dictionary.
- **No piece-selection algorithm from BitTorrent:** The random chunk selection (claimChunk) is a novel strategy optimized for local swarm dynamics, not BitTorrent's rarest-first heuristic.
- **No choking/unchoking algorithm:** The system uses bounded connection pools and peer-drop recovery rather than BitTorrent's tit-for-tat reciprocation.
- **Wi-Fi Direct cascading tree is wholly novel:** No existing P2P protocol addresses the single-radio constraint of mobile devices with a simultaneous receive-and-distribute tree topology.

5. Prototype

- User can select the firmware from internal/external storage on device
- Application continues parses and calculates metadata likes number of chunks, hash per-chunk, has per-firmware, name...

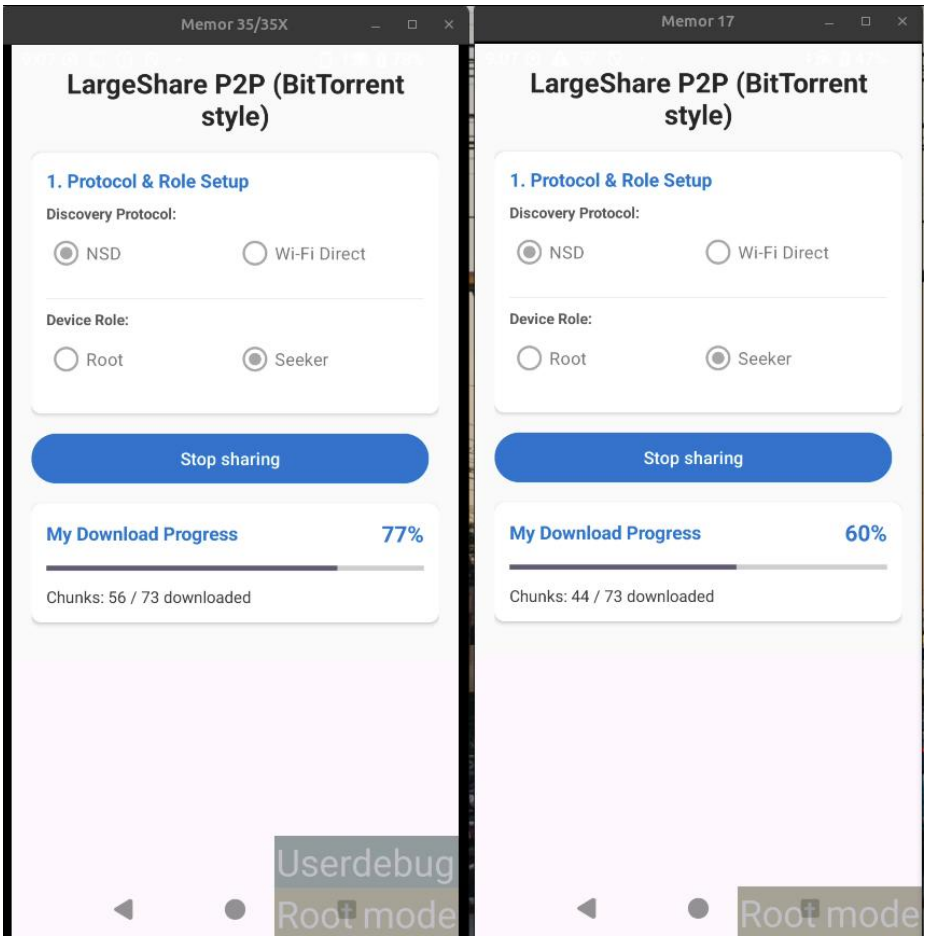


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Figure 7. Root device prepare metadata after downloading the image and start distributing the chunks

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- On each peer devices: start as seeker to request metadata from root device and fetch available chunks
- Application visualizes the process of seeking chunks



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- User can use root device to track the process of peers
- Application supports to filter out the peer by ID

